

#### 4.1 How do designer engineers use annotated 2D and 3D sketching and digital tools to graphically communicate ideas?

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| <p>a. Demonstrate an understanding of how to use annotated sketching and digital tools to graphically communicate ideas and sketch modelling to explore possible improvements, in terms of physical requirements, such as:</p> <ul style="list-style-type: none"> <li>• function, usability, construction, movement, stability, composition, strength</li> <li>• aesthetic qualities</li> <li>• manufacturing processes</li> <li>• suitability of materials and components.</li> </ul> |                                                                                     |
| <p>b. Demonstrate an understanding of methods used to represent systems and components to inform third parties, including:</p> <ol style="list-style-type: none"> <li>i. constructional diagrams/working drawings</li> <li>ii. digital visualisations</li> <li>iii. circuit and system diagrams</li> <li>iv. flowcharts with associated symbols</li> <li>v. prototypes and models.</li> </ol>                                                                                          |  |